

ILLEGAL PARKING DETECTION AND ENFORCEMENT ROUTE PRIORITIZATION



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1. INTRODUCTION

A vehicle parked in a prohibited roadside area can take up part of the road and disturb traffic. When several locations need to be checked, an enforcement officer also needs to decide which one to visit first. The closest location may be easy to reach, but another location may be causing a bigger obstruction.

This project proposes a system with two parts. Digital image processing (DIP) will examine fixed-camera footage and flag vehicles that stay in a marked no-parking area. reinforcement learning (RL) will then recommend which flagged location an officer should inspect next. Previous research shows why the system needs to distinguish a vehicle that briefly stops from one that remains stationary for longer [1].

The system is intended to recommend inspections, not automatically declare that a driver has broken a rule. Since this is still a proposal, the footage, annotations, and experimental results have not been collected yet.

2. PROBLEM AND SOLUTION

2.1 Problem

There are two main problems that need to be solved. First, the system needs to identify a possible illegal-parking event from video. a vehicle passing through a camera view should not create an alert. A vehicle that stops briefly should also be treated differently from one that remains in a no-parking area. The system needs to record how long the vehicle stays and estimate whether it occupies part of the roadway. Camera footage alone can't confirm every legal detail, so a person must review the alert.

Second, an officer may have several alerts at different locations but can only visit one at a time. visiting the nearest alert first could delay a more serious obstruction. Visiting the most serious alert first could require a long trip. The routing decision therefore needs to consider both the estimated obstruction and travel time.

2.2 Solution

For each camera view, we will manually mark two regions: the no-parking area and the part of the road that vehicles should keep clear. The image-processing system will compare video frames with an empty view of the same scene. It will then look for a vehicle-sized object that remains inside the marked area across multiple frames. Comparing a current image with a background image is an established approach in

parking-status research, although its performance in our roadside setting will still need to be tested [3]

Each candidate alert will contain its location, the time the vehicle was first observed, its observed duration, and obstruction score based on how much of the detected object overlaps the marked roadway region. The score is only an image-based estimate. It does not measure actual traffic delay.

A tabular Q-learning model will use the alerts and travel times between locations to recommend the next place to inspect. Research has used Q-learning in vehicle routing problems [4], but our specific next-location decision and reward design will need their own evaluation. The final prototype will display the annotated camera frame, the active alerts, and the recommended destination. No deep learning will be used.

3. METHOD

1. Collect and label video. We plan to use either a suitable public video dataset or footage recorded with a fixed camera where filming is allowed. Before selecting the source, we will check that the clips show moving vehicles, brief stops, and vehicles that remain stationary long enough to estimate their observed parking duration. We will mark the no-parking and roadway regions and label when each vehicle stops and leaves. Any missing labels will be added manually. Parking-space images alone can't serve as ground truth for illegal parking [2]
2. Prepare the images. We will keep the camera, keep the camera position fixed, resize frames to a consistent size, and adjust brightness if needed. An empty frame will be used as a reference for each camera view. We will compare later frames with that reference to locate objects that were not present in the empty scene. Changes in sunlight, shadows, and camera position may cause false detections, so these conditions will be included in testing
3. Create candidate alerts. We will threshold the difference image, use morphological opening and closing to clean the detected regions, and filter out objects with unsuitable sizes or shapes. A candidate must remain in the no-parking region for longer than a chosen duration threshold before it becomes an alert. We will track it across sampled frames to avoid treating a passing vehicle as parked. Research on vehicle stalls also shows the need to consider stops that may have a valid reason [1], so the alert will still require human review.

4. Train in the routing model. We will represent approximately three to five inspection locations as nodes in a small road network. The travel time between locations will be stored in a matrix. The Q-learning state will include the officer's current location, which alerts remain unresolved, and their severity levels. An action means choosing the next alert location. The reward will favor inspecting a more severe alert, while applying penalties for travel time and for leaving severe alerts waiting. Training will use simulated alert scenarios, separate scenarios will be kept for testing.
5. Evaluate the system. We will compare the DIP alerts with manually labeled video using precision, recall, and error in estimated parking duration. For routing, we will compare Q-learning against two simple rules: visiting the nearest alert first and visiting the most severe alert first. We will measure total travel time and how long alerts wait before inspection, with greater weight given to more severe alerts. The final output will include the DIP pipeline, the learned Q-table or policy, annotated example frames, and route recommendations.

4. REFERENCE

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